# EnergyBench: a live, gate-aligned benchmark of day-ahead forecasts for European power markets

Evgenii Genov[1] Ander Martinez Alonso[1] Christoph Bergmeir[2]

[1]EVERGi Research Group, Flanders Alliance for Climate Technology (FACT),
Vrije Universiteit Brussel, Pleinlaan 2, 1050 Brussels, Belgium

[2]Department of Computer Science and Artificial Intelligence, University of Granada, Granada, Spain

evgenii.genov@vub.be, Ander.Martinez.Alonso@vub.be, bergmeir@ugr.es

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energybench.foretec.co · github.com/ujohn33/foretec-energy-bench

**Abstract**

Comparisons of energy forecasting models usually rest on backtests, which are easy to get subtly wrong: the information available when a forecast would have been sold is replaced by whatever is convenient to download later. EnergyBench runs every entry live instead. Each day at 11:30 CET/CEST, thirty minutes before the day-ahead (SDAC) gate closes, every model forecasts the next day's price, wind and solar generation and load on the 15-minute grid for six bidding zones (BE, NL, FR, DE-LU, DK1, DK2). Unlike other live benchmarks, EnergyBench runs the models itself, on one information set that it freezes at the cut-off and publishes, including three numerical weather prediction models, so foundation models are compared with and without the same weather covariates under a real market deadline. Errors are pooled per series against a seasonal-naive baseline, and models are ranked by a Bradley–Terry rating on the Elo scale with a day-bootstrap interval. The field spans baselines, linear and gradient-boosted models on the frozen weather, eighteen time-series foundation models from eleven families, ensembles, and the grid operators' own forecasts. Over the first thirty live days, the covariates reduced the error of each of the ten foundation models run with and without them by 38–52% on matched races (45–52% for the seven that accept them natively; 64–72% on wind, 55–62% on solar). Used zero-shot, these models match or outperform a LightGBM model trained on the same covariates. Model size, from 3M to 331M parameters, matters far less, and median ensembles of the covariate models lead the standings. The grid operators' pre-gate forecasts remain ahead on wind and solar, where the best single foundation models have 20–28% more error; on load the two are level. A model enters the benchmark by pull request.

## 1 Introduction

Day-ahead forecasts of price, renewable generation and load are the inputs of every bidding decision on European power markets, and claims about how well a given model makes them are rarely comparable. Three things go wrong in the usual backtest. First, the information set drifts: a forecast nominally made at 11:30 is built from weather runs that were published hours later, from actuals that were still provisional, or from a TSO forecast that only appears after the gate. Second, the target may differ from the quantity the market trades, for example hourly values where the market now trades quarter-hours, generation definitions that differ between data providers, or actuals that were later revised. Third, the evaluation window, the baseline and the metric can be chosen after the results are known.

A fourth gap has opened with time-series foundation models. In operational energy forecasting, exogenous inputs are the standard, not an option: wind and solar forecasts are built on numerical weather predictions, load forecasts on temperature and calendar effects, and day-ahead price forecasts on forecasts of load, renewable infeed and fuel cost [8–10]. The grid operators' own forecasts, the industry reference, are NWP-driven, and the energy forecasting competitions supplied weather forecasts

as a matter of course [5]. The benchmarks on which foundation models are compared are, by contrast, mostly univariate. A model that forecasts tomorrow's solar output from the last four weeks of solar output alone is evaluated on a task that does not arise in practice, and a ranking obtained in this way is of limited use for choosing a model to deploy.

A live benchmark that provides every entry with the same inputs avoids the first two problems by construction. If each model runs at the actual cut-off on the data available at that time, weather forecasts included, and its forecast is locked before the actual gate, look-ahead is excluded, and the record accumulates day by day in public. The drawback is that a live record is short and its standings vary from day to day. The design described below accounts for this variability. Errors are pooled before they are normalised, rankings are reported with intervals, and only live forecasts are reported. The third problem, the choice of analysis after the results are known, is reduced but not removed by fixed public rules (Section 5).

EnergyBench went live on 2 October 2026 with three zones (BE, NL, FR) and three targets; total load was added on 6 October and three zones (DE-LU, DK1, DK2) on 15 October. The code and the configuration file that define the benchmark are versioned in a public repository. This paper describes that configuration as of 31 October 2026 and the results to date.

## 2 Related work

**Static benchmarks.** Forecasting competitions have shaped the field since the M-competitions [1]; the Monash archive [2], GIFT-Eval [3] and fev-bench [4] standardise the evaluation of general-purpose and foundation time-series models on fixed datasets. In energy, the Global Energy Forecasting Competitions [5] posed realistic tasks as one-off events, and Lago et al. [10] published an open benchmark for day-ahead price forecasting with fixed splits and day-ahead exogenous forecasts for five markets. Fixed datasets have two weaknesses for the present purpose. They describe a past regime, and energy systems change quickly; and their test periods may have been seen, directly or through pretraining corpora, by the models they evaluate. Retrospective energy studies also often leave the information set implicit: several price-forecasting studies use ENTSO-E's day-ahead wind and solar forecasts as inputs although those are published only after 18:00 on $D-1$, six hours after the auction closes [7].

**Covariates.** The value of weather inputs in energy forecasting is long established. Wind and solar power forecasting has been NWP-based since its beginnings, and the wind and solar tracks of GEFCom2014 gave every team ECMWF forecasts [5]; load forecasting rests on temperature, and the reviews of price forecasting single out forecasts of load and renewable generation as the most informative exogenous drivers [8–10]. Foundation models have only recently gained the means to use such inputs: Chronos-2, TimesFM, TiRex-2, Timestral, TimeH2O and the t0 models accept known-future covariates, by different mechanisms and with different success. Whether and how much they benefit is an empirical question that a univariate benchmark cannot answer, and it is the question an operational user needs answered first.

**Live benchmarks.** Two recent platforms evaluate forecasts made before the outcome exists, and each leaves open a question that EnergyBench is built to answer.

TS-Arena [6] implements a forecast pre-registration protocol for time-series foundation models: a round hands out each series' history up to its last published value and closes registration before the next value is recorded. Its 14 challenges draw on SMARD (German day-ahead price, load and generation), Gridstatus (US ISO hub prices, load and generation) and Fingrid (Finland), at 15-minute resolution with a one-day horizon and hourly with a three-day horizon, from a 1,000-step context. Models are scored by MASE and ranked by an online Elo rating whose interval comes from replaying the rounds in random order. Two features limit what its rankings say about energy forecasting. *First, there are no covariates*: the foundation models forecast wind, solar and load from the series' own history, without the weather forecast that any operational forecaster of those quantities would use, so the benchmark measures a task

no market participant faces, and a model's ability to exploit covariates, which in our results reduces the error of the same models by 38–52% (Section 4), is not measured at all. *Second, the information set is not tied to a market gate*: each round starts at the last published value, so the window follows each source's publication delay rather than the market. A German price forecast registered on day $X$ starts after the delivery day that cleared on $X$ and covers a later day than the next auction needs, and the generation and load windows do not align with a delivery day at all. What a forecaster could have known at a decision time is therefore not defined, and no forecast corresponds to a bid that could have been placed.

Energy-Arena [7] is an open, API-based platform whose challenges are defined by their market deadline: forecasts are submitted before 12:00 CET on $D-1$ for the whole next day, as point (RMSE), quantile (weighted interval score) or ensemble (CRPS) forecasts, with leaderboards over rolling windows. It is gate-aligned, but it compares *submissions*, not models under common conditions. Participants run their own models on any inputs they have, including proprietary weather, and describe them only if they choose to; they may also keep their forecasts hidden. Since the information set cannot be verified, a difference on the leaderboard may reflect better data rather than a better method, and entries cannot be reproduced. The platform names no reference models, so whether foundation models are represented at all depends on who submits; and its wind, solar and load challenges cover only Austria and Germany-Luxembourg (day-ahead prices extend to thirteen further zones). In industry, the Predico platform [14] scores live renewable forecasts submitted to Elia and 50Hertz against a real gate, for enrolled forecasters and without publishing their inputs.

**Positioning.** EnergyBench takes Energy-Arena's operational framing (the day-ahead gate defines the task) and TS-Arena's pre-registration logic (only a forecast made before the outcome exists is free of leakage), and closes the gaps between them by running every model itself. Each day it fetches one information set at the gate-aligned cut-off, freezes and publishes it, and gives it to every entry: the information set is enforced rather than declared, and any forecast can be reproduced from its archived inputs. Foundation models from seven families also run with the same frozen covariates, so the value of the weather is measured model by model rather than assumed away; all four targets are forecast in six zones; and the grid operators' own forecasts, before and after the gate, serve as references. The cost is that entries are submitted as code rather than as forecasts, and proprietary data cannot be used. Table 1 summarises the differences.

# 3 The benchmark

This section defines the benchmark as it runs: the forecasting task, the daily timeline and what a model may know at the cut-off, where the actuals come from, the covariates frozen for every entry, the field of entries, and how forecasts are scored, ranked and valued.

## 3.1 Task

For each zone $z \in \{\text{BE}, \text{NL}, \text{FR}, \text{DE-LU}, \text{DK1}, \text{DK2}\}$ and target $y \in \{\text{price}, \text{wind}, \text{solar}, \text{load}\}$ (24 series), a forecast made on issue day $D-1$ covers every quarter-hour of delivery day $D$ in local time: 96 values, or 92 and 100 on clock-change days. Price is the SDAC day-ahead price (EUR/MWh); wind is onshore plus offshore generation, solar is solar generation and load is total load (MW). Models return a point forecast and optionally the nine deciles $q_{0.1}, \ldots, q_{0.9}$. Danish load is published only hourly, so DK load forecasts are scored on hourly averages of forecast and actual: detail that the actual cannot resolve is neither rewarded nor penalised.

## 3.2 Timeline and information set

The benchmark day is defined by three clock times in CET/CEST (Figure 1). At the *cut-off*, 11:30 on $D-1$, a snapshot of everything public is taken and every model runs on it. The *gate* is 12:00, the SDAC gate closure. A live run that finishes after the gate is marked late and never scored, and a day's

Table 1: Live energy forecasting benchmarks compared, as described in their papers and on their websites (October 2026).

| | TS-Arena [6] | Energy-Arena [7] | EnergyBench (this paper) |
|---|---|---|---|
| Task timing | from each series' last published value; no market gate | submission before the 12:00 day-ahead gate, whole next day | 11:30 cut-off, locked before the 12:00 gate, whole next day |
| Zones and targets | 14 challenges (SMARD, Gridstatus, Fingrid) | wind, solar, load: AT, DE-LU; price: 15 zones | price, wind, solar, load in each of BE, NL, FR, DE-LU, DK1, DK2 |
| Who runs models | participants (containers) | participants (API submissions) | the benchmark, from submitted code |
| Information set | series history only | anything before the gate; self-declared, unverified | frozen and published: history, three NWP models, load forecast, fuel |
| Covariates for foundation models | none | if a participant adds them | eleven foundation models also with the same covariates |
| Foundation models | the focus of the field | none named; depends on submissions | eleven families, and ensembles |
| Transparency | platform code public | method description and forecasts optional | code, inputs and forecasts public |
| Scores | MASE | RMSE, WIS, CRPS | MAE, RMSE, bias, pinball; relative to seasonal naive, pooled per series |
| Ranking | online Elo, replay bootstrap | per metric over rolling windows | Bradley–Terry Elo, day bootstrap |
| References | statistical baselines | optional benchmark models | baselines; the TSOs' forecasts before and after the gate |

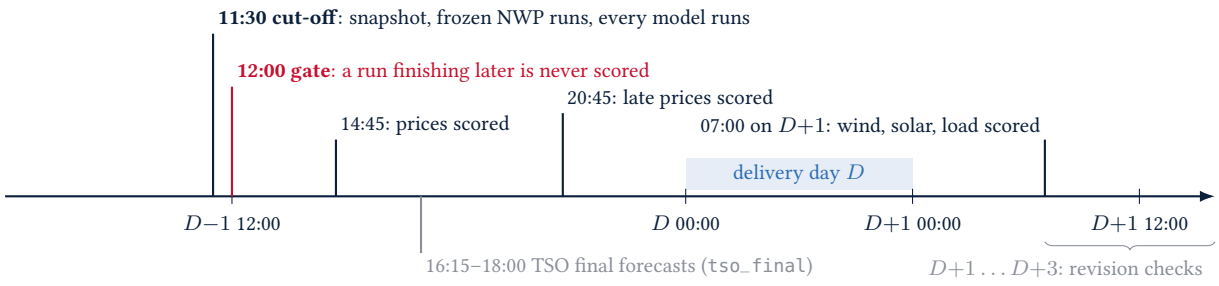



Figure 1: One benchmark day (not to scale). Everything a model sees is fixed at the 11:30 cut-off; the forecast must be locked before the 12:00 gate. The TSOs' final day-ahead wind and solar forecasts appear only after the gate and enter as references.

forecasts cannot be changed once written, so the record consists of the forecasts as they were at the gate. Prices are scored on $D-1$ after the auction for $D$ has cleared, at the scoring run of 14:45, or at that of 20:45 if the prices were not yet published. Wind, solar and load are scored at 07:00 on $D+1$ if every quarter-hour of $D$ has been published, and otherwise at the first later scoring run at which 95% has.

What a model may know at the cut-off is set per target. Prices: everything up to the end of $D-1$, which the auction on $D-2$ published. Wind, solar and load: actuals up to the cut-off minus one hour of publication lag. Every model receives the last 28 days of 15-minute history (the context). A model may train on longer history, provided that each training row uses only data from before that row's own cut-off. Two qualifications apply: the training actuals are the current values, which may have been revised since they were first published, and the weather in the two-year training frames of two entries is

matched to the live lead time rather than to the live run (Section 3.4). Training never takes place inside the live window. Models that retrain do so in the nightly catch-up at 03:30, and a live run that finds no trained model uses the latest earlier one instead of training. The TSOs' day-ahead renewable forecasts are not an input: Regulation (EU) 543/2013 [15] has them published by 18:00 on $D-1$, after the gate.

### 3.3 Actuals and provenance

The primary source is the ENTSO-E Transparency Platform [16], parsed from its XML: day-ahead prices (document A44, the 15-minute SDAC prices), actual generation per production type (A75; wind as onshore B19 plus offshore B18, solar B16) and actual total load (A65). ENTSO-E does not deliver every quarter-hour: alone it lacked 12% of Belgian and 2.6% of French wind quarter-hours in September 2026. Gaps are filled from the TSOs' own open data (Elia for Belgium, RTE éCO2mix for France), then from Energy-Charts as a last resort; where a TSO publishes finer data than it sends ENTSO-E, the TSO comes first (Elia sends hourly wind and solar to ENTSO-E but publishes quarter-hours). Every snapshot and every set of actuals records how many points came from each source, and the site shows it. Electricity Maps' choice of sources served as the reference for the new zones, with one exception found in operation: ENTSO-E's DK1 offshore wind covers only the parks registered there and has missed 0.3–0.5 GW of new offshore capacity since July 2026, so Danish wind has come from Energinet's real-time production data since delivery day 18 October (Danish solar and load stay on ENTSO-E). The first two live issue days, 2 and 3 October, were run and scored on Energy-Charts data alone, with wind and solar scored on $D+2$.

The Netherlands shows why provenance matters. ENTSO-E's Dutch series cover only what TenneT meters itself: solar at about 1% of the fleet (a capacity factor of 0.1% over 20 September–7 October 2026, against 10.7% in the national estimate, 12.7% in Belgium and 13.6% in France), wind mostly offshore, and load net of embedded solar (about 6 GW against 17 GW gross at a sunny midday). The Dutch wind, solar and load targets therefore come exclusively from NED, the Nationaal Energie Dashboard backed by TenneT and the distribution operators, which publishes national estimates: wind as onshore plus metered offshore, load as gross consumption as Elia and RTE report it. These series have no fallback source. The only other source, ENTSO-E, measures a different quantity (for solar, about 1% of the fleet), and filling a gap from it would combine two definitions within one delivery day; quarter-hours that NED has not published are therefore left unscored. Live forecasts made from the earlier ENTSO-E definition (issue days up to 10 October for wind and solar, 11 October for load) are kept and shown but never scored. When a series' definition changes but its earlier scores remain valid, the earlier actuals are locked: NED's offshore wind was a park-by-park model through delivery day 12 October, 7–27% above the metered output used since, and those days keep their scores.

Actuals are *frozen* the first time they are complete and scored from the frozen copy. Until $D+3$, wind, solar and load are re-fetched at every scoring run; a change above 0.5% of the day's energy re-freezes the actuals and re-scores the day, and every check is logged. Prices are final at the auction. Between the gate and scoring, the site shows the quarter-hours metered so far as provisional actuals, which are never scored.

### 3.4 Covariates frozen at the cut-off

Models that declare covariate inputs receive, per issue day, a table frozen at the cut-off and a log of what went into it.

**Weather.** Three independent global and regional NWP models, ICON-EU (DWD), GFS (NOAA) and IFS HRES (ECMWF), are fetched run by run from the Open-Meteo single-runs archive [17]. A run is admitted only if its initialisation is at most the cut-off minus a lag, the provider's measured publication delay plus a margin (3.5 h for ICON-EU, 7 h for GFS and IFS); at 11:30 this admits the 06 UTC ICON-EU run and the 00 UTC GFS and IFS runs. Since 19 October a run must also be complete: if any variable is less than 99% filled over lead times 12–48 h, which contain the delivery day, the fetch is retried up to five times and the builder then falls back to an earlier cycle, at most two (Section 5). Weather is sampled

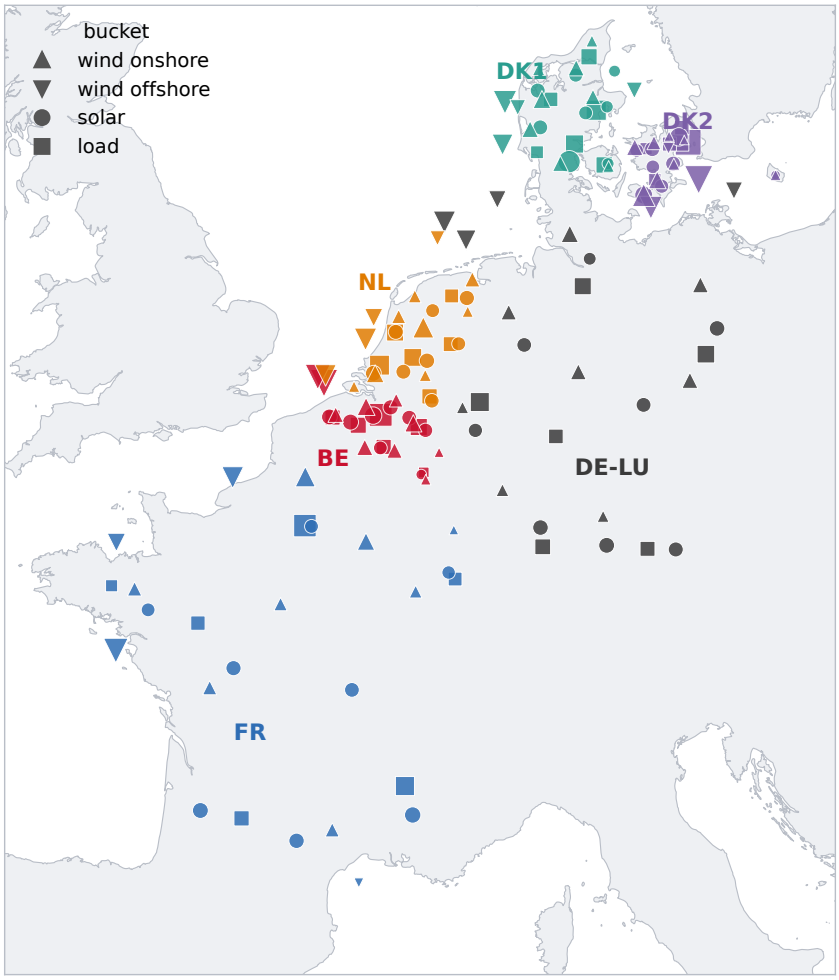



Figure 2: The 153 weather sampling points: capacity-weighted $k$-means centroids per zone and bucket, from public registers (load: population). Marker size is the point's share of its zone and bucket.

at 153 capacity-weighted $k$-means centroids in all, per zone 8 for onshore wind, 8 for solar, 6 for load (weighted by population) and 2–4 for offshore wind. They are built from public data: OpenStreetMap turbine positions scaled to Elia's capacity and municipal solar data of the distribution operators for BE, RIVM turbine positions and CBS municipal solar capacity for NL, ODRÉ's national register of installations at commune level for FR, the Marktstammdatenregister for DE-LU and Energinet's capacity per municipality for DK1 and DK2. The table carries, per variable and bucket, each NWP model's capacity-weighted zone mean, and across the three models the ensemble mean, spread and pairwise differences, so that disagreement can serve as a feature. Models trained on the 28-day context train on tables built by the same rule day by day, from the runs that were available at each day's own cut-off. The two-year training frames of `lgbm_prod` and `lin_cov_2y` predate the single-runs archive for ICON-EU and GFS; there the archive's previous-runs product is matched to the live lead time to the nearest day, which gives forecasts of the same lead but not always of the same run (Section 5). Figure 2 shows the sampling points.

**Load forecast.** ENTSO-E's day-ahead total load forecast (A65), which TSOs must publish two hours before the gate. It is an input of the price models; load models also received it until issue day 15 October and no longer do, because in most zones it is itself the pre-gate reference they are compared with (Section 4.7). Section 5 records a parsing fault in its Danish series.

**Fuel cost.** Daily closes of TTF gas (ICE Dutch front month) and of an exchange-traded EU carbon allowance certificate, as published by Yahoo Finance, the last close dated two days before delivery or earlier, and the derived short-run marginal cost of a gas plant, $\mathrm{TTF}/0.52 + 0.37\,\mathrm{EUA}$ (EUR/MWh).

### 3.5 Entries

A model is a folder with a declarative `model.yaml`: a named sktime [18] estimator, a Python module, or a subprocess script with its own environment, the route all foundation models take (each model loads its weights once per run, for all series). A model whose environment is missing is skipped and logged, not failed. The live run takes 10–20 minutes for the field on a four-core, 8 GB server (17 minutes for 24 series on 15 October), against a 30-minute window; from 16 October two models run concurrently, each with half the CPU threads. A model is deployed by adding its folder to `models/`. The live run executes every enabled folder, and the nightly catch-up runs each new model over a fixed backtest window (14 September – 1 October 2026, shown on the site as a separate record and not used in this paper). The field on 31 October (Table 2) has 43 active entries.

No TSO in France or Germany publishes a day-ahead wind or solar forecast before the gate (RTE at 16:15, the German TSOs at 18:00 on $D-1$), and Energinet's is not verified as published before it, so `tso_cutoff` has wind and solar only for BE and NL. Classical univariate models (ARIMA, ETS, Theta, CES, TBATS) were in the field in its first two weeks and were retired: on 15-minute data with a 96-step daily cycle and a 28-day context they were not competitive. Their records are kept but no longer shown.

### 3.6 Scoring

For every forecast, issue day $d$ and series $s$ the benchmark records MAE, RMSE, bias and, for models with deciles, the weighted interval score, the Winkler score and coverage of the 80% interval, and the pinball loss averaged over the nine deciles,

$$\mathrm{PL} = \frac{1}{9\,T}\sum_{t=1}^{T}\sum_{\tau\in\{0.1,\ldots,0.9\}}\max\big(\tau\,(y_t - q_{\tau,t}),\,(\tau-1)(y_t - q_{\tau,t})\big), \tag{1}$$

with $T$ the scored quarter-hours of the day. Errors are compared across series of different scale by relating them to the baseline `naive_weekly`, and they are aggregated *before* they are normalised: for model $m$, series $s$ and the set $\mathcal{D}_{m,s}$ of issue days on which both $m$ and the baseline were scored,

$$\mathrm{RelMAE}_{m,s} = \frac{\sum_{d\in\mathcal{D}_{m,s}}\sum_t |e^{m}_{d,t}|}{\sum_{d\in\mathcal{D}_{m,s}}\sum_t |e^{\mathrm{base}}_{d,t}|}, \qquad \mathrm{RelRMSE}_{m,s} = \Big(\frac{\sum_{d,t}(e^{m}_{d,t})^2}{\sum_{d,t}(e^{\mathrm{base}}_{d,t})^2}\Big)^{1/2}. \tag{2}$$

A ratio of means rather than a mean of daily ratios keeps one calm day, on which the baseline happened to be nearly right, from dominating the average. Over several series the per-series values are combined by the mean, the median or the geometric mean; the site lets the reader switch. The geometric mean, used in the tables below, treats halving and doubling the error symmetrically and penalises a model that fails on one series.

Three choices in these definitions affect which model is ranked best, and they are stated here explicitly. *The functional rewarded.* Absolute error is minimised by the median of the predictive distribution and squared error by its mean [11]; for skewed targets such as solar power or spiking prices the two differ, and a model trained on quantile loss is not wrong to forecast the median. The field mixes both: most foundation models report the median of their predictive quantiles (Chronos, TiRex, Moirai, Sundial, Toto, t0, Timestral, Tabby), while TimesFM and the LightGBM and ridge entries, fitted on squared error, report a conditional mean, as does the climatology. The standings are built on MAE and so reward the median; Rel. RMSE is shown next to it so that mean-oriented entries are not judged on the other functional only, and bias is reported separately because neither accuracy measure distinguishes a systematic offset from noise. *The benchmark in the denominator.* Every scale-free error measure is defined relative to

Table 2: The field, 31 October 2026 (43 active entries). Parameter counts are those of the checkpoints the benchmark loads. Covariates: *native*, the model's own known-future covariate input; *ridge*, an in-context ridge regression on the covariates whose residual the model forecasts; *full*, the whole covariate table.

| Entry | Params | Covariates | Developer, set-up |
|---|---|---|---|
| *Baselines* | | | |
| `naive_weekly` | | | same quarter-hour one week earlier; the reference of all relative errors |
| `naive_daily` | | | same quarter-hour one day earlier |
| `climatology` | | | same local time of day within $\pm 14$ days, 2023 to August 2026 |
| `lin_lags`, `lgbm_lags` | | | ridge and LightGBM on the series' own lags; two years, retrained monthly |
| *Models trained on the covariates*[a] | | | |
| `lin_cov`, `lgbm_cov` | | full | ridge and LightGBM on the 28-day context |
| `lin_cov_2y` | | full | ridge on two years |
| `lgbm_prod` | | full | LightGBM on two years, 120-day recency half-life, solar on its capacity factor[b] |
| *Foundation models, zero-shot*[c] | | | |
| Chronos-2 [19] | 120M | native | Amazon |
| Chronos-Bolt base [19] | 205M | | Amazon |
| TimesFM 2.5 [20] | 231M | ridge | Google |
| TimesFM 3.0 [20] | 331M | native | Google |
| Moirai 2 small [21] | 11M | [d] | Salesforce |
| TiRex [22] | 35M | | NX-AI |
| TiRex-2 [22] | 95M | native | NX-AI; about 38M active in univariate mode |
| Toto | 151M | [d] | Datadog; retired 27 October, superseded by Toto 2 |
| Toto 2 | 313M, 1B | | Datadog; live from 25 October |
| Sundial | 128M | | Tsinghua University |
| TTM r2 | 3.4M | | IBM |
| Tabby, Tabby-Pretrain | 147M, 146M | ridge | Huawei Noah's Ark Lab; load forecast without covariates |
| Timestral | 209M | native | Mistral AI |
| TimeH2O | 108M | native | H2O.ai; live from 28 October, with covariates from 29 October |
| t0-alpha, t0-beta, t0-gamma | 102M, 256M, 256M | native | The Forecasting Company |
| *Ensembles: quarter-hourly median of fixed members, point and each quantile; live from 20 October* | | | |
| `ens_fm` | | | the newest history-only model of five native-covariate families |
| `ens_fm_cov` | | native | their five covariate variants |
| `ens_hybrid` | | native | those five and `lgbm_prod` |
| *References: the grid operators' forecasts* | | | |
| `tso_cutoff` | | | as published before the gate; ranked; live only[e] |
| `tso_final` | | | final day-ahead wind and solar, published after the gate; never ranked[f] |

[a] *Full*: every NWP model's columns, the ensemble statistics and differences, the calendar and, for price, yesterday's price. The foundation models take a compact subset instead: three or four ensemble means and spreads per target, and for price the load forecast, fuel cost and the renewable drivers of the coupled zones.

[b] Retrained daily for price and load, monthly for wind and solar.

[c] Every foundation model runs on the series' own history; those with a covariate entry run a second time with the frozen covariates (e.g. `chronos2` and `chronos2_cov`).

[d] Covariate variants retired after trial runs: covariates made Moirai 2 worse, and Toto gained too little for its run time and memory.

[e] Elia's 11:00 day-ahead wind and solar forecast for BE, or, where that field is still empty at the run, Elia's most recent forecast read at the run; NED's national wind, solar and load forecast for NL, fetched at the cut-off; the ENTSO-E day-ahead load forecast elsewhere.

[f] Elia 18:00, RTE 16:15, ENTSO-E for DE-LU and Danish wind, NED in the evening for NL.

a benchmark forecast [12]. EnergyBench uses the value of the same quarter-hour one week earlier, a seasonal naive forecast that reproduces the daily and weekly cycles of price and load and requires no fitting. It is evaluated on the same test days as the model rather than in-sample, which is possible because every series has 96 scored values per day [13]. For wind and solar this benchmark is weak, and relative errors well below one are to be expected. It serves as a common unit of comparison and is not a measure of how difficult a series is to forecast. *Aggregation.* Errors are averaged over the 96 quarter-hours of the delivery day and over issue days within a series, and then across series with equal weight per series. With equal weights, the standings measure performance across forecasting problems rather than where most energy or revenue is at stake; the forecast-value view (Section 3.8) addresses the latter for wind and solar.

### 3.7 Ranking

A *race* is one issue day and one series; its field is every model with a scored forecast in it. A model whose run failed is left out of that race rather than ranked last (the run tracker on the site shows every failure), and a model joins the races of the series it declares from the day it is added. Leaving failures out is defensible only because the benchmark runs every entry itself: an entry cannot choose the days on which it forecasts. A benchmark that accepted submitted forecasts would have to count a missing forecast as a loss, or replace it by the baseline, since a participant could otherwise withhold forecasts on difficult days. Within a race every pair of models plays one game, won by the lower relative error of the day (equal errors split the game). All games of the selected period are fitted at once with a Bradley–Terry model [23], $P(i \text{ beats } j) = p_i/(p_i + p_j)$, by the minorisation–maximisation algorithm of Hunter [24], and reported on the Elo scale, $\mathrm{Elo}_i = 1000 + 400 \log_{10} p_i$, with the geometric mean of the strengths fixed at 1 so that 1000 is the field's average. One virtual drawn game against an average opponent keeps the rating finite for a model that wins or loses every game. Unlike a running Elo, the rating does not depend on the order of the races, and a win over a strong model counts more than a win over a weak one; the same construction underlies the Chatbot Arena leaderboard [25].

The 95% interval is the 2.5th–97.5th percentile of 200 refits on issue days drawn with replacement. Days rather than races are resampled because all races of a day share the same weather and tend to be won and lost together. Overlapping intervals mean the order of two models is not settled. Any pairwise rating is relative to the field: a model's score depends on whom it played, and several near-identical entries (three releases of one model family, say) take wins from each other and from everyone else. The pooled relative errors do not depend on the field and are always reported next to the rating. A model is listed as provisional if it has fewer races than the smaller of 24 and half the most any model has. The standings can be restricted to the last 7 or 30 days, to a zone or a target, and to models with or without covariates; filters choose which races count and which models are shown, never how a race is scored.

### 3.8 Forecast value

Accuracy does not translate directly into revenue. For wind and solar, the benchmark also values each forecast as the day-ahead income of a price-taking producer that owns a virtual 100 MW asset in each zone, producing like the zone's whole fleet: actual output and every forecast are scaled by 100 MW over the installed capacity, so that every zone counts equally. Before the gate the producer sells a fixed percentile of the forecast distribution (or the point forecast) at the day-ahead price. Each deviation is settled at the zone's published imbalance price (ENTSO-E document 17.1.G; the German price is the national reBAP). By default only the error cost counts: a deviation of size $\delta$ costs $|\delta| \cdot |\pi^{\mathrm{imb}} - \pi^{\mathrm{DA}}|$, with the imbalance price of its own direction, so no error ever earns and the score, revenue as a share of perfect-foresight revenue, is at most 100%. The alternative settles deviations as each zone would pay them, two prices where the TSO publishes two (NL and FR) and one elsewhere; there a deviation can earn when the imbalance price happens to sit on its side, which over a few weeks rests on a handful of price spikes. The producer cannot curtail and does not trade intraday.

Table 3: Scored issue days per series in the live record, 2–31 October 2026.

| | BE | NL | FR | DE-LU | DK1 | DK2 |
|---|---|---|---|---|---|---|
| Price | 30 | 30 | 30 | 17 | 17 | 17 |
| Wind | 30 | 21 | 30 | 17 | 14 | 14 |
| Solar | 30 | 21 | 30 | 17 | 17 | 17 |
| Load | 26 | 20 | 26 | 17 | – | – |

### 3.9 Operation and reproducibility

The benchmark is a set of systemd timers on one server: the live run at 11:30, scoring at 07:00, 14:45 and 20:45, an hourly fetch of provisional actuals, and a catch-up at 03:30. Every forecast, input snapshot, covariate table, provenance file and revision log is kept, and the website is a static export of them, so any number on it can be recomputed. The configuration file that defines the benchmark (cut-off, gate, resolution, deciles, context, baseline, backtest window, sources, admission lags) is versioned with the code: changing it changes the benchmark. On the site, backtest and live results are always shown apart; this paper reports live results only.

## 4 Results: the first thirty live days

The results below are those of the live record only, scored on the forecasts as they were locked before each gate. After the coverage of the record, they give the standings, the value of the covariates for foundation models, the role of model size, and the comparison with purpose-built models, with the full field over its last twelve days and with the grid operators' own forecasts.

### 4.1 Coverage

The live record covers issue days 2–31 October 2026, the last day with every target scored when this paper was written. It is uneven by design: series enter the record when they can be scored on a stable definition, and not before (Table 3). BE, NL and FR price, wind and solar run from 2 October and load from 6 October; the Dutch wind, solar and load forecasts made before the switch to NED's national series (Section 3.3) are voided, so the Dutch record starts on 11–12 October. DE-LU, DK1 and DK2 run from 15 October. Danish wind lost two issue days to its own switch of source. Danish load has been published by ENTSO-E only in part since 10 October, which leaves nine scored days, and the pre-gate reference covered only part of the delivery day on ten zone-days (Section 5). A series with so few days would weigh as much as any other in the averages below, so Danish load is left out. The record therefore holds 22 of the 24 series, with between 14 and 30 issue days each. Models also joined at different times, which is why every comparison below is made on the races the compared models share.

### 4.2 Standings

Table 4 gives the standings for the 40 entries with at least 100 races, and Figure 3 shows their ratings. The two median ensembles of covariate models, live since 20 October, rate highest (`ens_hybrid`, Elo 1270 and Rel. MAE 0.233; `ens_fm_cov`, 1228 and 0.242). Among single models, t0-gamma, Timestral, TimesFM 3.0 and Chronos-2 with covariates lead (Elo 1185–1208, Rel. MAE 0.273–0.297), with overlapping intervals. Behind the leading group the field splits in two. Every covariate-aware foundation model except TimesFM 2.5 is separated by its interval from every history-only model, and the best history-only entries, Toto 2 at 1B parameters over its seven days and the history-only ensemble (Elo 1000 and 978), rank behind fifteen covariate models. Among the history-only foundation models the differences are small (pooled Rel. MAE 0.53–0.79, leaving out Toto 2) compared with the effect of giving any of them the weather (0.27–0.37 with covariates, Table 5).

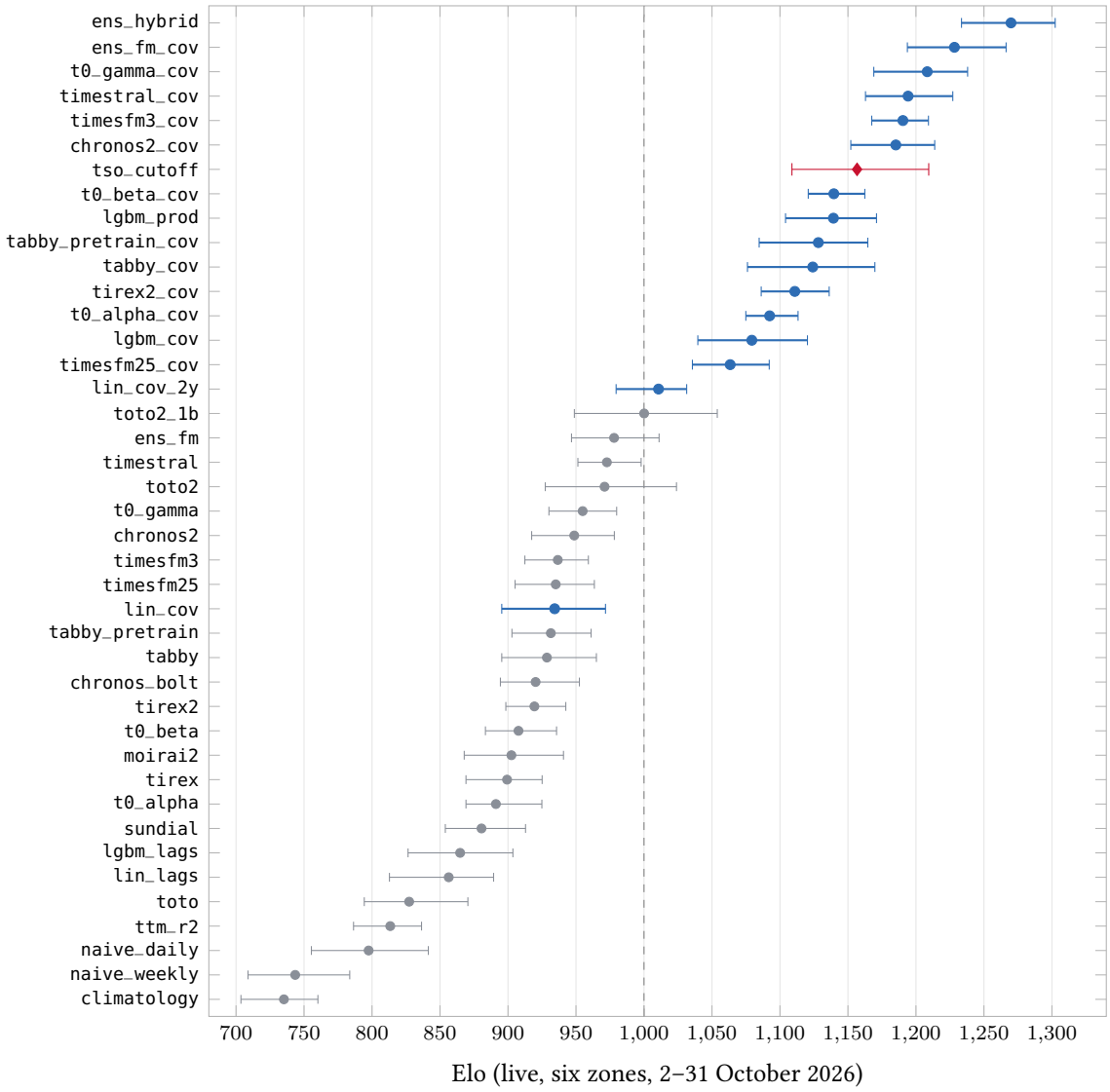



Figure 3: Live Elo ratings with day-bootstrap 95% intervals, entries of Table 4. Blue: models using the frozen covariates; grey: history only; red diamond: the pre-gate TSO forecast `tso_cutoff`, a reference rated on 8 of the 22 series only. The dashed line is the field average.

The grid operators' pre-gate forecast (`tso_cutoff`) is listed below the ranked entries, as a reference. It covers only 8 of the 22 series, so its rating rests on games in those series alone, and its pooled error is a mean over a different set of series, weighted towards load, than that of the entries that forecast all 22. Neither number can be compared directly with those of the ranked entries. A rating also counts wins rather than margins. On the 93 races they share, `ens_hybrid` has the lower error in 48% of them, and its pooled error is level with that of the operators (Section 4.7), although its rating is about 110 points higher. Section 4.7 therefore compares the operators with each entry on the races both forecast.

Figure 4 breaks the pooled error down by series. The covariate models lead on wind and solar in every zone, by a margin much larger than the differences between them; Dutch and Danish solar are the hardest of these series. On load the history-only foundation models are level with the covariate ones, and the grid operators' pre-gate load forecast is strong only in France.

Table 4: Live standings, issue days 2–31 October 2026, all six zones (22 series). Elo with its day-bootstrap 95% interval; Rel. MAE and Rel. RMSE pooled per series against `naive_weekly`, geometric mean over the series a model forecasts ("all") or over the zones of one target. Entries with at least 100 races; covariate models in bold.

| | Model | Elo | 95% interval | Races | Rel. MAE | Rel. RMSE | Price | Wind | Solar | Load |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | all series | | Rel. MAE per target | | | |
| 1 | **ens_hybrid** | 1270 | [1234, 1302] | 262 | 0.233 | 0.265 | 0.202 | 0.128 | 0.290 | 0.513 |
| 2 | **ens_fm_cov** | 1228 | [1194, 1266] | 262 | 0.242 | 0.275 | 0.203 | 0.135 | 0.302 | 0.534 |
| 3 | **t0_gamma_cov** | 1208 | [1169, 1238] | 346 | 0.274 | 0.307 | 0.234 | 0.165 | 0.323 | 0.574 |
| 4 | **timestral_cov** | 1194 | [1163, 1227] | 346 | 0.273 | 0.303 | 0.261 | 0.152 | 0.305 | 0.590 |
| 5 | **timesfm3_cov** | 1190 | [1168, 1209] | 481 | 0.283 | 0.306 | 0.237 | 0.163 | 0.332 | 0.663 |
| 6 | **chronos2_cov** | 1185 | [1152, 1214] | 481 | 0.297 | 0.328 | 0.238 | 0.186 | 0.353 | 0.642 |
| 7 | **t0_beta_cov** | 1140 | [1121, 1162] | 467 | 0.322 | 0.357 | 0.275 | 0.197 | 0.388 | 0.647 |
| 8 | **lgbm_prod** | 1139 | [1104, 1171] | 346 | 0.293 | 0.309 | 0.271 | 0.141 | 0.365 | 0.702 |
| 9 | **tabby_pretrain_cov** | 1128 | [1085, 1165] | 346 | 0.313 | 0.339 | 0.275 | 0.194 | 0.358 | 0.638 |
| 10 | **tabby_cov** | 1124 | [1076, 1170] | 346 | 0.314 | 0.339 | 0.272 | 0.196 | 0.367 | 0.630 |
| 11 | **tirex2_cov** | 1111 | [1086, 1136] | 442 | 0.312 | 0.329 | 0.271 | 0.165 | 0.388 | 0.722 |
| 12 | **t0_alpha_cov** | 1092 | [1075, 1113] | 467 | 0.337 | 0.358 | 0.346 | 0.173 | 0.378 | 0.740 |
| 13 | **lgbm_cov** | 1079 | [1040, 1120] | 481 | 0.339 | 0.353 | 0.317 | 0.156 | 0.376 | 1.035 |
| 14 | **timesfm25_cov** | 1063 | [1036, 1092] | 481 | 0.367 | 0.392 | 0.319 | 0.199 | 0.405 | 0.981 |
| 15 | **lin_cov_2y** | 1011 | [979, 1031] | 380 | 0.396 | 0.408 | 0.290 | 0.186 | 0.453 | 1.609 |
| 16 | toto2_1b | 1000 | [949, 1054] | 154 | 0.485 | 0.543 | 0.275 | 0.415 | 0.822 | 0.652 |
| 17 | ens_fm | 978 | [947, 1011] | 262 | 0.495 | 0.528 | 0.336 | 0.489 | 0.679 | 0.564 |
| 18 | timestral | 973 | [951, 998] | 346 | 0.528 | 0.556 | 0.351 | 0.552 | 0.695 | 0.603 |
| 19 | toto2 | 971 | [927, 1024] | 154 | 0.498 | 0.556 | 0.289 | 0.406 | 0.849 | 0.691 |
| 20 | t0_gamma | 955 | [930, 980] | 346 | 0.543 | 0.572 | 0.355 | 0.535 | 0.741 | 0.662 |
| 21 | chronos2 | 949 | [917, 978] | 488 | 0.586 | 0.617 | 0.380 | 0.564 | 0.884 | 0.645 |
| 22 | timesfm3 | 937 | [912, 959] | 481 | 0.595 | 0.620 | 0.373 | 0.590 | 0.880 | 0.675 |
| 23 | timesfm25 | 935 | [905, 964] | 481 | 0.592 | 0.625 | 0.383 | 0.554 | 0.827 | 0.760 |
| 24 | **lin_cov** | 934 | [895, 972] | 380 | 0.464 | 0.480 | 0.388 | 0.207 | 0.575 | 1.469 |
| 25 | tabby_pretrain | 932 | [903, 961] | 346 | 0.563 | 0.591 | 0.382 | 0.589 | 0.731 | 0.638 |
| 26 | tabby | 929 | [895, 965] | 346 | 0.571 | 0.599 | 0.373 | 0.586 | 0.798 | 0.630 |
| 27 | chronos_bolt | 920 | [894, 953] | 481 | 0.607 | 0.636 | 0.387 | 0.575 | 0.914 | 0.700 |
| 28 | tirex2 | 919 | [898, 942] | 442 | 0.586 | 0.606 | 0.388 | 0.535 | 0.866 | 0.694 |
| 29 | t0_beta | 908 | [883, 936] | 467 | 0.601 | 0.625 | 0.412 | 0.546 | 0.899 | 0.670 |
| 30 | moirai2 | 902 | [868, 941] | 481 | 0.618 | 0.651 | 0.413 | 0.547 | 0.915 | 0.758 |
| 31 | tirex | 899 | [869, 925] | 481 | 0.621 | 0.644 | 0.407 | 0.575 | 0.832 | 0.845 |
| 32 | t0_alpha | 891 | [869, 925] | 467 | 0.614 | 0.637 | 0.408 | 0.570 | 0.860 | 0.766 |
| 33 | sundial | 880 | [854, 913] | 481 | 0.636 | 0.641 | 0.430 | 0.590 | 0.942 | 0.713 |
| 34 | lgbm_lags | 865 | [827, 904] | 488 | 0.687 | 0.709 | 0.497 | 0.580 | 0.927 | 0.919 |
| 35 | lin_lags | 856 | [813, 889] | 380 | 0.628 | 0.615 | 0.453 | 0.504 | 0.912 | 0.815 |
| 36 | toto | 827 | [794, 871] | 393 | 0.790 | 0.813 | 0.556 | 0.763 | 0.966 | 1.040 |
| 37 | ttm_r2 | 813 | [786, 836] | 481 | 0.692 | 0.673 | 0.455 | 0.529 | 0.929 | 1.245 |
| 38 | naive_daily | 797 | [755, 841] | 488 | 0.825 | 0.877 | 0.517 | 0.634 | 0.995 | 1.855 |
| 39 | naive_weekly | 744 | [709, 784] | 488 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| 40 | climatology | 735 | [704, 760] | 481 | 1.007 | 0.961 | 1.147 | 0.649 | 0.901 | 1.895 |
| *References: the grid operators' forecasts (Section 4.7)* | | | | | | | | | | |
| – | tso_cutoff | 1157 | [1109, 1209] | 182 | 0.374 | 0.390 | – | 0.139 | 0.350 | 0.634 |
| – | tso_final | – | – | – | 0.175 | 0.193 | – | 0.115 | 0.267 | – |

Races differ because entries joined at different times (Toto 2, 25 October; ensembles 20 October; t0-gamma, Timestral, Tabby, `lgbm_prod` 16 October) or left (Toto, 27 October); Table 6 compares the whole field on common days. The references are not ranked. `tso_cutoff` forecasts 8 of the 22 series (BE and NL wind and solar; load in BE, NL, FR and DE-LU), so its Elo and pooled errors are not comparable with those of the ranked entries (Section 4). `tso_final`, published after the gate, is shown for wind and solar in BE, NL, FR and DE-LU; its Danish wind forecasts cover a different scope from the actuals and are left out (Section 4.7). TimeH2O, live from 28 October, has too few races to list.

### 4.3 Covariates roughly halve the error of foundation models

Ten foundation models run both on the series' own history and with the frozen covariate table, so the value of the weather can be measured model by model on exactly the same races (Table 5; TimeH2O is too recent to include). All ten gain from the covariates. For the seven that take covariates natively, the pooled relative error falls by 45–52%. The reduction is 64–72% on wind and 55–62% on solar, where a

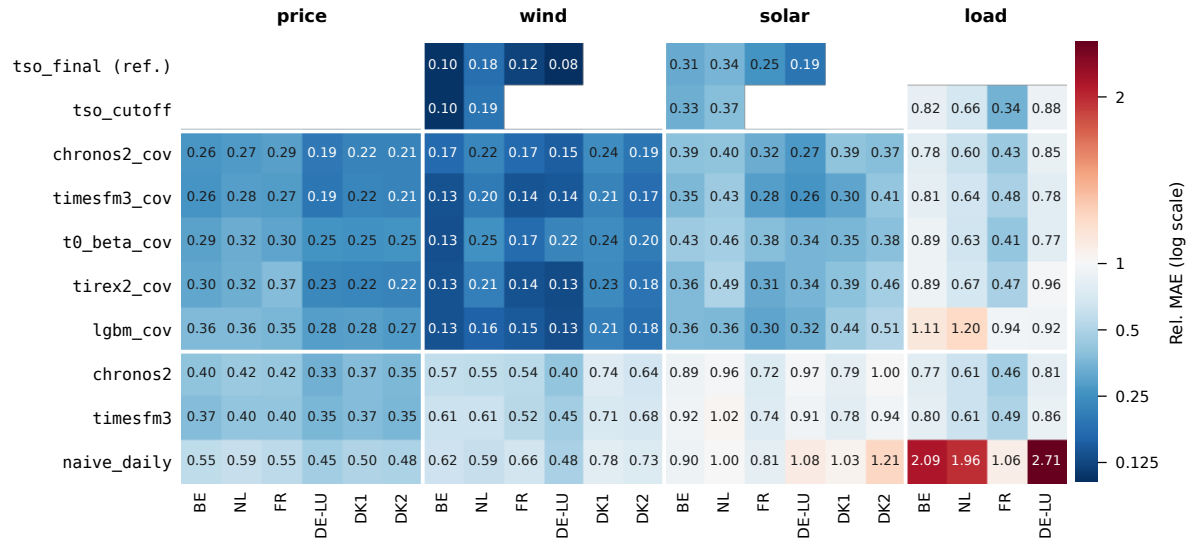



Figure 4: Pooled Rel. MAE per series for the references, five covariate models, two history-only foundation models and `naive_daily` (live, 2–31 October 2026; colour on a log scale, 1 is the seasonal-naive baseline). Empty cells: no forecast, or (for `tso_final` in DK1 and DK2) a forecast of a different scope.

Table 5: The same foundation model without and with the frozen covariates, on the races both forecast (live, all zones, issue days 2–31 October 2026). "Native": the model takes known-future covariates itself; "ridge": an in-context ridge regression on the covariates, the model forecasting its residual (Tabby forecasts load without covariates). Rel. MAE pooled per series, geometric mean over series; a change is the geometric mean of the per-series ratios minus one.

| | | | | Rel. MAE | | Change with covariates | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Model | Covariates | Days | Races | without | with | all | price | wind | solar | load |
| Chronos-2 | native | 29 | 481 | 0.585 | 0.297 | −49% | −37% | −67% | −60% | +0% |
| TimesFM 3.0 | native | 29 | 481 | 0.595 | 0.283 | −52% | −36% | −72% | −62% | −2% |
| t0-alpha | native | 27 | 467 | 0.614 | 0.337 | −45% | −15% | −70% | −56% | −3% |
| t0-beta | native | 27 | 467 | 0.601 | 0.322 | −46% | −33% | −64% | −57% | −3% |
| t0-gamma | native | 16 | 346 | 0.543 | 0.274 | −50% | −34% | −69% | −56% | −13% |
| TiRex-2 | native | 24 | 442 | 0.586 | 0.312 | −47% | −30% | −69% | −55% | +4% |
| Timestral | native | 16 | 346 | 0.528 | 0.273 | −48% | −26% | −72% | −56% | −2% |
| TimesFM 2.5 | ridge | 29 | 481 | 0.592 | 0.367 | −38% | −17% | −64% | −51% | +29% |
| Tabby | ridge | 16 | 346 | 0.571 | 0.314 | −45% | −27% | −67% | −54% | +0% |
| Tabby-Pretrain | ridge | 16 | 346 | 0.563 | 0.313 | −44% | −28% | −67% | −51% | +0% |

model that sees only the last four weeks of output has no information about tomorrow's wind or cloud cover, and 15–37% on price, through the load forecast, fuel cost and the expected renewable infeed. On load the change is small (−13% to +4%), since a load curve is dominated by daily and weekly cycles that the history already contains. Of the three models that receive the covariates through an in-context ridge regression, the two Tabby models gain nearly as much (44–45%). TimesFM 2.5 gains less (38%), and its load error rises by 29%, because the ridge regression fitted on 28 days extrapolates poorly.

The effect does not depend on the zone (Figure 5, where every point of a zone is computed on the same races). In every zone the covariates reduce Chronos-2's error by 50–67% on wind and solar and by 31–42% on price, while on load they change it by between −6% and +5%. On wind and solar the grid operators' forecasts are ahead of Chronos-2 with covariates in every zone where they exist, and `lgbm_cov` beats both of them in NL; on load, Chronos-2 with covariates is ahead of the pre-gate forecast in BE, NL and DE-LU and behind it in FR.

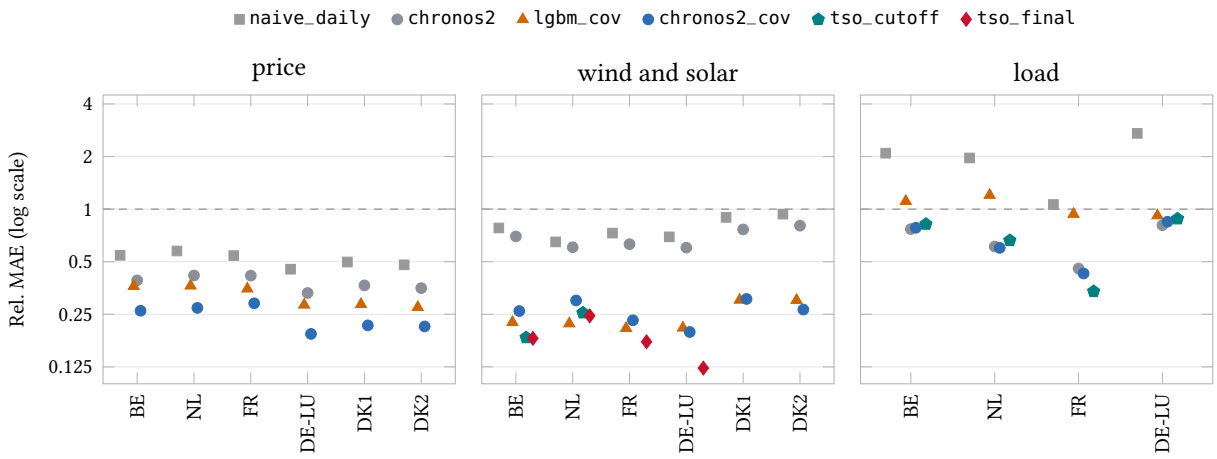



Figure 5: Pooled Rel. MAE per zone and target group (live, 2–31 October 2026, log scale; the dashed line is the seasonal-naive baseline). Every point of a zone and panel is computed on the same races: those that all models shown there forecast, the TSO references included where they exist (`tso_cutoff`: BE and NL wind and solar, and load; `tso_final`: wind and solar outside Denmark, Section 4.7). Danish load is left out (Section 4).

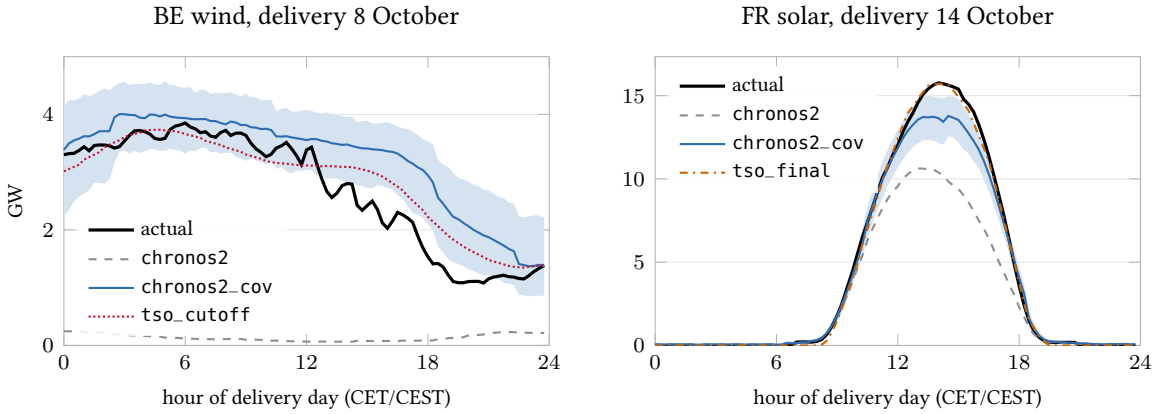



Figure 6: Two delivery days: actual output, Chronos-2 without and with covariates (shaded: the covariate version's 10–90% band), and a TSO forecast (Elia's pre-gate forecast for BE; RTE's final forecast, published after the gate, for FR).

Figure 6 illustrates the difference on two delivery days on which the covariates helped more than in the median race. In the median wind or solar race, Chronos-2 with covariates has 41% of the error of Chronos-2 without; on these two days the shares are 21% and 28%. On 8 October Belgian wind generation was between 3.1 and 3.9 GW until midday and fell to about 1.1 GW by the evening. Chronos-2 on its own history carried its recent output forward and forecast less than 0.3 GW for the whole day. The covariate version followed the forecast weather, although, like Elia's pre-gate forecast, it anticipated the evening decline too late. On 14 October French solar peaked at 15.8 GW: Chronos-2 alone reached 10.6 GW, the covariate version 13.8 GW, and RTE's final forecast, published after the gate, 15.7 GW.

## 4.4 Model size

Figure 7 sets each foundation model's error against its number of parameters, on the sixteen issue days 16–31 October, on which fifteen of them ran (346 races each; Toto left the field on 27 October, and Toto 2 and TimeH2O joined after the 16th). Without covariates, larger models do somewhat better (Spearman $\rho = -0.52$, $p = 0.05$), but the spread is small and not monotone: Timestral (209M), t0-gamma (256M)

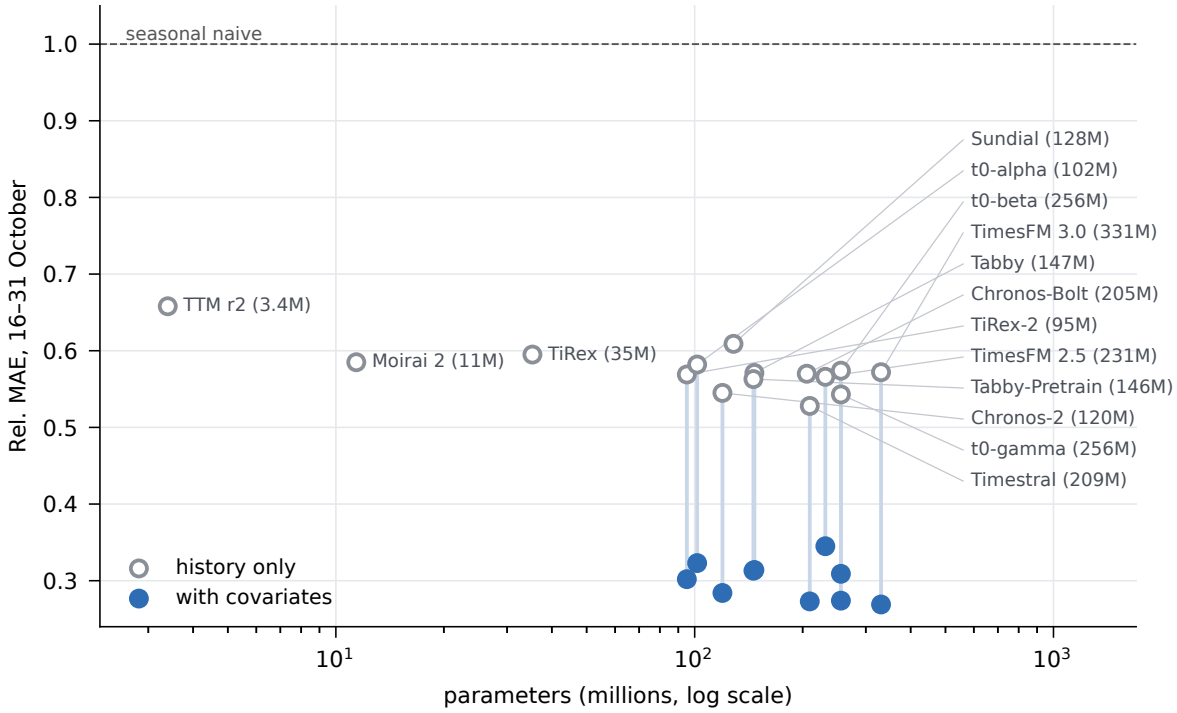



Figure 7: Pooled Rel. MAE against parameter count for fifteen foundation models, without covariates (open) and, where a variant exists, with the frozen covariates (filled, joined to the same model without), on the issue days 16–31 October 2026 (all zones, 346 races per model). Parameter counts are those of the checkpoints the benchmark loads; TiRex-2 uses about 38M of its 95M in univariate mode.

and Chronos-2 (120M) lead the rest by at least 0.02, TimesFM 3.0, the largest at 331M, sits mid-field, and TTM r2 at 3.4M is the weakest. Above about 100M the relation disappears ($\rho = -0.19$, 11 models), and it is not significant with covariates ($\rho = -0.38$, ten models), where TimesFM 3.0, Timestral and t0-gamma lead Chronos-2 by 0.01–0.02. Every covariate model, whatever its size, has a lower error than every history-only model, so the weather input matters far more than any difference in size.

### 4.5 Against purpose-built models

The covariate-aware foundation models are used zero-shot, yet of the five that take covariates natively and have run over almost the whole record, four have a lower error than a LightGBM model trained on the same covariate table, on the races each shares with it (`lgbm_cov`, Rel. MAE 0.331–0.339, against 0.283–0.322), and t0-alpha is level with it (0.337 against 0.338). LightGBM receives more of the table: every NWP model's columns, the calendar and, for price, yesterday's price, whereas the foundation models receive a small set of ensemble means and spreads. LightGBM is the best of them on wind (0.154–0.156, against 0.163–0.197) and level on solar (0.357–0.376, against 0.332–0.388), but far behind on load (1.025–1.035), where 28 days are too few to learn the response of load to the weather. Trained on two years, `lgbm_prod` does better. On the days both ran (16–31 October) it reaches 0.293, against 0.322 for `lgbm_cov`, within the range of the native-covariate foundation models (0.269–0.323).

### 4.6 The full field over the last twelve days

Nine entries joined on 16 October (t0-gamma, Timestral, Tabby and Tabby-Pretrain, each with and without covariates, and `lgbm_prod`) and the three ensembles on 20 October. Table 6 compares the field on the twelve issue days 20–31 October, on which all of them ran (262 races each); Toto 2 and TimeH2O joined later and Toto left. The median of the five covariate foundation models (`ens_fm_cov`, Rel. MAE

Table 6: The field on the issue days 20–31 October 2026, all zones (262 races per entry). The fifteen covariate entries, the three best history-only entries and the baseline, with ranks among all entries that ran on every one of these days; below the line the pre-gate TSO forecast as an unranked reference (8 series, 93 races). Columns as in Table 4.

| | Model | Elo | 95% interval | Rel. MAE | Price | Wind | Solar | Load |
|---|---|---|---|---|---|---|---|---|
| 1 | **ens_hybrid** | 1272 | [1242, 1312] | 0.233 | 0.202 | 0.128 | 0.290 | 0.513 |
| 2 | **ens_fm_cov** | 1230 | [1199, 1267] | 0.242 | 0.203 | 0.135 | 0.302 | 0.534 |
| 3 | **t0_gamma_cov** | 1213 | [1181, 1259] | 0.254 | 0.214 | 0.142 | 0.334 | 0.525 |
| 4 | **timesfm3_cov** | 1202 | [1167, 1235] | 0.246 | 0.196 | 0.138 | 0.308 | 0.587 |
| 5 | **timestral_cov** | 1197 | [1165, 1239] | 0.253 | 0.245 | 0.130 | 0.307 | 0.537 |
| 6 | **lgbm_prod** | 1170 | [1144, 1201] | 0.255 | 0.235 | 0.113 | 0.331 | 0.661 |
| 7 | **chronos2_cov** | 1163 | [1129, 1206] | 0.273 | 0.214 | 0.171 | 0.326 | 0.602 |
| 8 | **tabby_cov** | 1133 | [1082, 1195] | 0.290 | 0.238 | 0.177 | 0.363 | 0.580 |
| 9 | **tabby_pretrain_cov** | 1133 | [1086, 1189] | 0.291 | 0.242 | 0.175 | 0.359 | 0.599 |
| 10 | **t0_beta_cov** | 1131 | [1108, 1167] | 0.292 | 0.244 | 0.175 | 0.368 | 0.583 |
| 11 | **tirex2_cov** | 1120 | [1090, 1161] | 0.276 | 0.229 | 0.133 | 0.388 | 0.654 |
| 12 | **lgbm_cov** | 1095 | [1049, 1145] | 0.293 | 0.270 | 0.122 | 0.342 | 0.986 |
| 13 | **t0_alpha_cov** | 1089 | [1062, 1114] | 0.304 | 0.307 | 0.147 | 0.369 | 0.662 |
| 14 | **timesfm25_cov** | 1073 | [1034, 1119] | 0.316 | 0.245 | 0.177 | 0.381 | 0.824 |
| 15 | **lin_cov_2y** | 1019 | [988, 1048] | 0.354 | 0.277 | 0.162 | 0.411 | 1.308 |
| 16 | ens_fm | 978 | [949, 1006] | 0.495 | 0.336 | 0.489 | 0.679 | 0.564 |
| 17 | timestral | 966 | [940, 996] | 0.495 | 0.338 | 0.512 | 0.642 | 0.562 |
| 18 | t0_gamma | 951 | [922, 980] | 0.511 | 0.340 | 0.498 | 0.705 | 0.604 |
| 37 | naive_weekly | 699 | [647, 743] | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| – | tso_cutoff (ref.) | 1163 | [1090, 1235] | 0.312 | – | 0.087 | 0.329 | 0.576 |

0.242) has a lower error than each of its members (0.246–0.276). With lgbm_prod, a model of a different class trained on the same weather, as a sixth member, the median (ens_hybrid, 0.233) has a lower error than every single model. An ensemble is fixed in advance and does not require knowing which member will perform best, which makes it the more practical choice, although twelve days are too few to rank it reliably.

### 4.7 Against the grid operators

The grid operators' forecasts are compared with each entry on exactly the races both forecast. Two references are used. The pre-gate forecast (tso_cutoff) is the operators' forecast as published before the gate, and is therefore information that a market participant could have used. The final forecast (tso_final) is published between 16:15 and 18:00 on $D-1$, after the gate. It is based on later weather runs and more recent measurements than any entry can use, and serves as an upper reference rather than as a competitor.

Table 7 sets the foundation models with covariates against tso_cutoff on the same 125 races: the sixteen issue days 16–31 October, BE and NL wind and solar, and load in BE, NL, FR and DE-LU. The operators remain ahead overall. The closest single models, Timestral and t0-gamma, have 10% and 12% more error (95% day-bootstrap intervals $-2$ to $+23$ and $-1$ to $+27$). TimesFM 2.5, which takes the covariates through a ridge regression and gains least from them (Table 5), has 63% more. Within a model family the newer release has the lower error on these races: TimesFM 3.0 has 27% less error than TimesFM 2.5 (interval 22–31) and t0-gamma 16% less than t0-alpha (8–22). All of these models ran over the same weeks, so these are comparisons between models, not trends over time.

The gap differs by target. On load the closest models are level with the operators overall (Timestral $+0.4$%, t0-gamma $-2$%): they have 7–16% less error in BE, NL and DE-LU, and 30–45% more in FR. On wind and solar they have 20–28% more error. There lgbm_prod, trained on two years of these series, comes closest (4% more error, interval $-7$ to $+14$), but it has 19% more error on load. Over the twelve days on which the ensembles ran (93 races), ens_hybrid is level with the operators overall ($+2$%, interval $-11$ to $+18$), with 17% more error on wind and solar and 11% less on load. On wind and solar most of the improvement over a history-only model comes from the weather input. On the races of Table 7, TimesFM 3.0 reaches 0.56 of the baseline's error on wind from its history alone and 0.16 with

Table 7: Foundation models with covariates and `lgbm_prod` against the pre-gate TSO forecast `tso_cutoff`, on the same 125 races (issue days 16–31 October 2026; BE and NL wind and solar, load in BE, NL, FR and DE-LU): extra pooled error, geometric mean over series, for all eight series with its day-bootstrap 95% interval and per target group. Negative: the entry has the lower error.

| Model | All | 95% interval | Wind, solar | Load |
|---|---|---|---|---|
| Timestral | +10% | [−2, +23] | +20% | 0% |
| t0-gamma | +12% | [−1, +27] | +28% | −2% |
| TimesFM 3.0 | +20% | [+7, +34] | +29% | +11% |
| Chronos-2 | +20% | [+4, +38] | +40% | +3% |
| t0-beta | +24% | [+9, +41] | +42% | +7% |
| TiRex-2 | +29% | [+16, +44] | +36% | +23% |
| Tabby | +30% | [+14, +49] | +57% | +7% |
| t0-alpha | +33% | [+18, +48] | +42% | +23% |
| TimesFM 2.5 | +63% | [+48, +80] | +62% | +65% |
| `lgbm_prod` | +11% | [+1, +22] | +4% | +19% |

covariates, against 0.12 for the operators; on solar the values are 0.89 and 0.39, against 0.29. On load it barely changes, from 0.66 to 0.65, against 0.59. Outside NL the pre-gate load reference is the ENTSO-E load forecast, which the load models received as an input until 15 October.

The final forecasts, published after the gate, are further ahead. On wind and solar in BE, NL, FR and DE-LU (90 races each) they reach Rel. MAE 0.115 and 0.268, against 0.138 for `lgbm_cov` on wind and 0.327 for TimesFM 3.0 on solar, the best entries on those races. Over its 16 days `lgbm_prod` comes closest on wind (0.119, against 0.111). Danish wind is left out of this comparison because Energinet's final forecast and the actuals do not measure the same quantity: its day-ahead forecast for DK1 wind has run at about 0.8 of measured output for two years, a difference in level rather than in skill.

## 5 Limitations

The record is short and covers one season; the standings will move. The weather inputs are three deterministic models served by one provider, and their spread stands in for forecast uncertainty. That provider is a single point of failure: on six of the thirty live days (3, 4, 5, 10, 13 and 17 October) it served the 06 UTC ICON-EU run at the cut-off before the run was fully ingested, with missing values over the delivery day. Models built on the three-model ensemble means were barely affected; models that read each NWP model's own columns lost accuracy on those days. A completeness check has rejected such runs since 19 October; the affected days are scored as they were run. The two-year training frames match the lead time of the live runs but take ICON-EU and GFS from a previous-runs product that can contain runs initialised after the cut-off of the day they describe; this concerns the training data of two entries, never what is scored. On seven issue days in DK1 and three in DK2, Energinet's load forecast on ENTSO-E ended at about 11:00 on the delivery day when the run read it, with a closing zero that was parsed as zero load for the rest of the day. Those `tso_cutoff` forecasts are voided, and the later hours, published after the gate, cannot repair them; until the parser was fixed on 31 October the same zeros reached the load forecast input of the price models. Those forecasts are scored as they were run: Danish price errors show no visible harm on those days, but the counterfactual was not run. The entries' settings were chosen before they went live, but by people who had seen the weeks before. The covariate models are deliberately simple and untuned; `lgbm_cov` and `lin_cov` train on the 28-day context for comparability with the foundation models, a known handicap that `lgbm_prod` removes. The Dutch targets changed definition during the record, and the weather points rest on public capacity registers of uneven detail (municipal data in BE, NL and DK, commune centres in FR, unit positions in DE-LU). The `tso_cutoff` reference exists only where a TSO publishes before the gate (BE and NL wind and solar, and load), and the Danish final forecasts measure a different scope from the actuals. The three zones

added on 15 October have seventeen issue days, and entries that joined late are compared with the rest only on the days they share; both will mean more with time. A model that fails is left out of the races it missed (Section 3.7), so a model that fails often is flagged by the run tracker rather than penalised in the rating. The value computation assumes a price-taking producer without intraday trading or curtailment.

The forecasts are fixed before the outcome is known, but the analyses in Section 4 were not registered in advance. Their windows follow from the dates on which entries joined, and the authors chose the comparisons with the record in view. This is the third problem of the introduction, in reduced form: the scores, forecasts and analysis scripts are public, so each comparison can be repeated on other windows, but only a set of comparisons fixed before the data arrive would remove the freedom entirely. Finally, the results describe only the entries in the field.

## 6 Conclusion

A live benchmark aligned with the market gate removes the freedom to choose a favourable backtest. EnergyBench runs one daily for the European day-ahead market, on quarter-hours, from frozen public data with recorded provenance, with ranking statistics that state their uncertainty and a value view that connects accuracy to revenue. After thirty live days in six zones, the main finding is that the weather information available at the cut-off matters more than the model that uses it. The same foundation model forecasts with about half the error when it is given that information (for Chronos-2, in every zone). Zero-shot foundation models with covariates match or outperform a gradient-boosting model trained on the same covariate table, and the median of several such models has a lower error than any of them. Model size matters far less. The grid operators' pre-gate forecasts remain ahead on wind and solar, where the best single foundation models have 20–28% more error, while on load the two are level; the operators' final forecasts, published after the gate, are further ahead. The repository accepts entries by pull request, and further zones, references and settlement rules will be added in the same way.

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